Tutorial letter 101/0/2026

INTRODUCTION TO DISCRETE MATHEMATICS
MAT2612

Year module

Department of Mathematical Sciences

IMPORTANT INFORMATION

Please register on *my*Unisa, activate your myLife e-mail account and make sure that you have regular access to the *my*Unisa module website, MAT2612-26-Y, as well as your group website.

Note: This is a fully online module. It is therefore, only available on *my*Unisa.

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1 INTRODUCTION

Dear Students

Unisa is a comprehensive open distance e-learning (CODeL) higher education institution. Our comprehensive curricula encapsulates a range of offerings, from strictly vocational to strictly academic certificates, diplomas and degrees. Unisa's openness and its distance eLearning character result in many students registering at Unisa who may not have had an opportunity to enrol in higher education. Our CODeL character implies that our programmes are carefully planned and structured to ensure success for students ranging from the under-prepared but with potential to the sufficiently prepared.

Multiple modes of delivery are involved in teaching and learning in a CODeL context, ranging from blended to fully online learning. As a default position, all postgraduate programmes are offered fully online with no printed study materials. Undergraduate programmes are offered using a blended mode of delivery where printed study materials are augmented with online teaching and learning via myUnisa, our learning management system. In some instances, undergraduate programmes are offered fully online as well.

Furthermore, our programmes are aligned with the vision, mission and values of the University. Unisa's commitment to serving humanity and shaping futures – combined with a clear appreciation of our location on the African continent – means that Unisa's graduates have distinctive graduate qualities, which include:

- Being independent, resilient, responsible and caring citizens able to fulfil and serve in multiple roles in their immediate and future local, national and global communities.
- Having a critical understanding of their location on the African continent with its histories, challenges and potential in relation to globally diverse contexts.
- The ability to critically analyse and evaluate the credibility and usefulness of information and data from multiple sources in a globalised world with ever-increasing information and data flows and competing worldviews.
- Knowing how to apply their discipline-specific knowledges competently, ethically and creatively to solve real-life problems.
- An awareness of their own learning and developmental needs and future potential.

Welcome to the MAT2612 module. We trust that you will find it both interesting and rewarding. This module is offered fully online, meaning that all information is available via the internet, we use *myUnisa* platform as our virtual campus. This is an online system that is used to administer, document and deliver educational material to you and support engagement with you. Look out for information from your lecturer as well as other Unisa platforms to determine how to access the virtual *myUnisa* module site. Information on the tools that will be available to engage with the lecturer and fellow students to support your learning will also be communicated via various platforms.

You are encouraged to log into the module site (MAT2612-26-Y) on *my*Unisa regularly (that is, at least twice per week). This tutorial letter contains important information about the scheme of work and resources for this module as well as exam admission. We urge you to read it carefully before working through the study material, preparing the assignment(s), preparing for the examination and addressing questions to your lecturers.

In this tutorial letter, you might find the instructions on the preparation and submission of the assignments. This tutorial letter also provides all the information you need with regard to the prescribed study material and other resources. Please study this information carefully and make sure that you obtain the prescribed material as soon as possible.

You will access all files online, a number of tutorial letters for example, solutions to assignments (feedback to assignments can also be discussed on MS Teams), during the year. These tutorial letters will be uploaded on myUnisa, under **Offical Study materials** or **Additional Resources** and **Lessons** tools on myUnisa platform. A tutorial letter is our way of communicating with you about teaching, learning and assessment.

Right from the start we would like to point out that **you must read all the tutorial letters** you access from the module site **immediately and carefully**, as they always contain important and, sometimes urgent information.

Because this is a fully online module, you will need to use *my*Unisa to study and complete the learning activities for this course. Please visit the website for MAT2612 on *my*Unisa frequently. The website for your module is MAT2612-26-Y.

We wish you every success with your studies!

2 MODULE OVERVIEW

2.1 Purpose

This module forms part of various qualifications and introduces students to the fundamental concepts of discrete mathematics. It focuses on propositional logic, proof techniques (including mathematical induction), counting methods such as permutations, combinations, and the pigeonhole principle, as well as relations, digraphs, functions, and ordered structures like lattices and Boolean algebras. Discrete mathematics traditionally addresses mathematical problems that arise in areas such as computer science (including aspects of logic), operations research (e.g. combinatorial optimization), and biotechnology (in particular the study of the human genome). Students credited with this module will be able to reason about discrete structures, construct mathematical arguments, and apply combinatorial methods to solve problems involving sets of discrete objects.

2.2 Outcomes

The broad outcomes for this module are to understand, compute and apply the following discrete mathematical concepts:

2.2.1 Mathematical Logic

- Prove identities/equalities/inequalities as well as other properties which hold for each of a set indexed by the natural numbers, by induction.

2.2.2 Counting

- Solve a variety of counting problems using the multiplication principle, sum rule, and the five basic formulas: arrangement with unlimited repetition, arrangement with fixed repetition, arrangement with no repetition, choice with no repetition, and choice with unlimited repetition.
- Solve problems using the pigeonhole principle.

2.2.3 Relations and Digraphs

- Draw the directed graph and set up the matrix of a finite relation.
- Decide whether or not a relation has various properties such as reflexivity, irreflexivity, symmetry, asymmetry, antisymmentry, transitivity by using the definition of the relation, the matrix or the set of pairs.
- Give paths and in/out degrees for a relation.
- Use Warshall's algorithm to find the transitive closure of a relation.
- Decide whether a given relation is an equivalence relation and if so, to give the equivalence classes.
- Find the complement and compositions of given relations.

2.2.4 Functions

- Decide whether a given function is: everywhere defined, one-to-one, onto, bijective, invertible and to find the inverse when the function is invertible.
- Use special functions such as the log function, ceiling function, base 2 exponential function and examine their properties.
- Use the definition of $O(f)$ to decide whether g is $O(f)$ for given functions f, g.
- Write a permutation as a product of disjoint cycles and as a product of transpositions.
- Decide whether a given permutation is even or odd.
- Find compositions and inverses of permutations.

2.2.5 Order Relations and Structures

- Decide, with reasons, whether a given set and relation define a partial order.
- Draw the Hasse diagram of a partial order.
- Decide whether or not two partial orders are isomorphic.
- Identify maximal elements in Hasse diagrams or explain why they do not exist.
- Decide, with reasons, whether a given partially ordered set is a lattice.
- Decide whether two lattices are isomorphic.
- Decide, with reasons, whether a given partially ordered set is a Boolean Algebra.
- Set up truth tables and Karnaugh maps for Boolean functions.

3 CURRICULUM TRANSFORMATION

Unisa has implemented a transformation charter that places curriculum transformation high on the teaching and learning agenda. Curriculum transformation includes student-centred scholarship, the pedagogical renewal of teaching and assessment practices, the scholarship of teaching and learning, and the infusion of African epistemologies and philosophies. All of these are being phased in at both programme and module levels.

As a result of this, you will notice a marked change in the teaching and learning strategy implemented by Unisa, together with the way in which the content is conceptualised in your modules. We encourage you to embrace these changes during your studies at Unisa, responsively and within the framework of transformation.

4 LECTURER(S) AND CONTACT DETAILS

4.1 Lecturer(s)

The primary lecturer for this module is:

Mr M Kgarose

Department: Mathematical Sciences
Telephone: +27 11 670 9169
Email: kgarom@unisa.ac.za

The secondary lecturer for this module is:

Dr Y Sithole
Department: Mathematical Sciences
Telephone: +27 11 670 9147
Email: sithoy@unisa.ac.za

A notice will be posted on *myUnisa* if there are any changes and/or an additional lecturer is appointed to this module.

Please do not hesitate to consult your lecturer whenever you experience difficulties with your studies. You may contact your lecturer by phone or through correspondence or by making a personal visit to his/her office. **Please arrange an appointment in advance (by telephone or by e-mail) to ensure that your lecturer will be available when you arrive.** Please come to these appointments well prepared with specific questions that indicate your own efforts to have understood the basic concepts involved. If these difficulties concern exercises which you are unable to solve, you must send us your attempts so that we can see where you are going wrong.

If you should experience any problems with the exercises in the study guide or prescribed book, your lecturer will gladly help you with them, provided that you send in your bonafide attempts. **When sending in any queries or problems, please do so separately from your assignments and address them directly to your lecturer.**

4.2 Department

You can contact the Department of Mathematical Sciences as follows:

Department of Mathematical Sciences

Departmental Secretary:
Telephone: 011 670 9147 (RSA) +27 11 670 9147 (International)
Email: swanemm@unisa.ac.za or mathsciences@unisa.ac.za

4.3 University

Contact addresses of the various administrative departments appear on the Unisa website:

http://www.unisa.ac.za/sites/corporate/default/Contact-us/Student-enquiries

Please include your student number in all correspondence.

5 RESOURCES

5.1 Prescribed book(s)

The prescribed book for this module is

Title:	**Discrete Mathematical Structures**
Author:	B. Kolman, R.C. Busby & S. Ross
Edition:	**6th Edition**, Pearson New International Edition
Year:	2014
Print Book ISBN10:	1-292-02484-4
Print Book ISBN13:	978-1-29202-484-4

You may also download and use the electronic version of this book from the Unisa Library website:

Please buy the textbook as soon as possible since you have to study from it directly – you cannot do this module without the prescribed textbook. Please refer to the list of official booksellers and their addresses in the *Study @ Unisa* brochure. Prescribed books can be obtained from the University's official booksellers. If you have difficulty in locating your book(s) at these booksellers, please contact the Prescribed Book Section at telephone number +27 11 429 4152 or e-mail vospresc@unisa.ac.za.

5.2 Recommended book(s)

There are no recommended books for this module.

5.3 Electronic reserves (e-Reserves)

E-reserves can be downloaded from the Library catalog. More information is available at http://oasis.unisa.ac.za/search/r.

5.4 Library services and resources

The Unisa Library offers a range of information services and resources and has made numerous library guides available at http://libguides.unisa.ac.za.

Recommended guides

- For brief information on the library, go to https://www.unisa.ac.za/library/libatglance
- For more detailed library information, go to http://www.unisa.ac.za/sites/corporate/default/Library
- For Frequently Asked Questions, go to https://www.unisa.ac.za/sites/corporate/default/Library/Frequently-Asked-Questions
- For research support and services such as the Personal Librarian service and the literature search request (on your research topic) service, offered by the Information Search Librarian, go to http://www.unisa.ac.za/sites/corporate/default/Library/Library-services/Research-support
- For library training for undergraduate students, go to https://www.unisa.ac.za/sites/corporate/default/Library/Library-services/Training
- For lending services, go to https://www.unisa.ac.za/sites/corporate/default/Library/Library-services/Lending-services
- For services for Postgraduate students, go to https://www.unisa.ac.za/sites/corporate/default/Library/Services-for-Postgraduates
- For support and services for students with disabilities, go to https://www.unisa.ac.za/sites/corporate/default/Library/Services-for-students-with-special-needs
- For Library technology Support, go to https://libguides.unisa.ac.za/techsupport
- For information on finding and using library resources and tools, go to http://libguides.unisa.ac.za/Research_skills
- For an A–Z list of library databases, go to https://libguides.unisa.ac.za/az.php

Important contact information

- Technical problems encountered in accessing library online services: Lib-help@unisa.ac.za
- General library-related queries: Library-enquiries@unisa.ac.za
- Queries related to library fines and payments: Library-fines@unisa.ac.za
- Interlibrary loan service for postgraduate students: libr-ill@unisa.ac.za
- Literature Search Service: Lib-search@unisa.ac.za

- Services and resources available to clients living with disabilities: lib-disability@unisa.ac.za
- Library book requests from, and book deliveries to Correctional Services: lib-corrections@unisa.ac.za
- Social media channels: Facebook: Unisa Library and X (Twitter): @UnisaLibrary

To view the Library orientation video – please click here:
Unisa Library and Information Services Video_1_1 (2).mp4

6 STUDENT SUPPORT SERVICES

6.1 Study @ Unisa publication and myModules

The *Study@Unisa* brochure is available on *my*Unisa: www.unisa.ac.za/brochures/studies

This brochure contains important information and guidelines for successful studies through Unisa.

If you need assistance with regard to the myModules system, you are welcome to use the following contact details:

- Toll-free landline: 0800 00 1870 (Select option 07 for myModules)
- Email: mymodule22@unisa.ac.za or myUnisaHelp@unisa.ac.za

You can access and view short videos on topics such as how to view your calendar, how to access module content, how to view announcements for modules, how to submit assessment and how to participate in forum activities via the following link:
https://dtls-qa.unisa.ac.za/course/view.php?id=32130

Registered Unisa students receive a free myLife e-mail account. Important information, notices and updates are sent exclusively to this account.

Please claim your e-mail account immediately after registering at Unisa by following this link: myLifeHelp@unisa.ac.za. Note that it can take up to 24 hours for your account to be activated after you have claimed it.

Your myLife account is the only e-mail account recognised by Unisa for official correspondence between you and the University and it will remain your official primary e-mail address on record at Unisa.

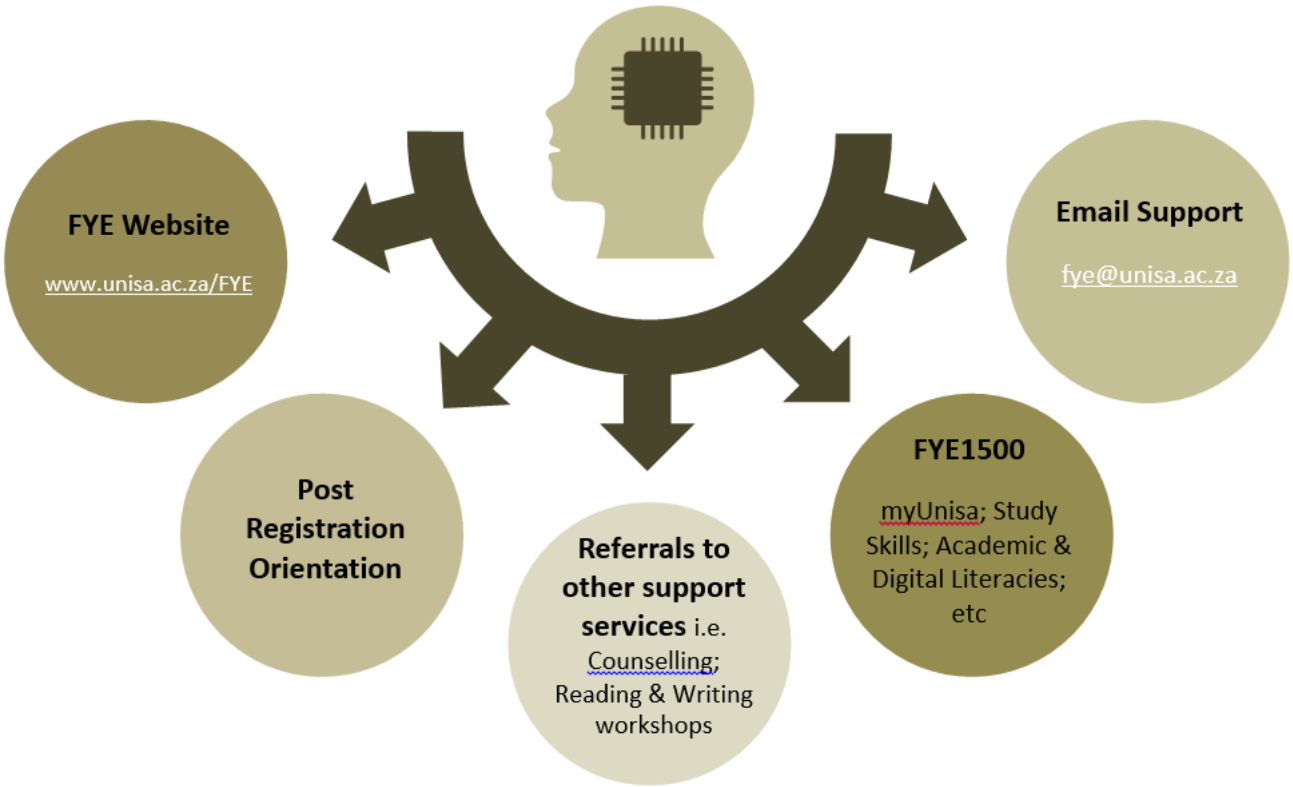
You remain responsible for the management of this e-mail account.

6.2 The Unisa First-Year Experience Programme

Many students find the transition from school education to tertiary education stressful and this is often true for students enrolling at Unisa for the first time. Unisa is a dedicated open distance and e-learning institution and is very different from face-to-face/contact institutions. It is a mega university, and all its programmes are offered through either blended learning or fully online learning. For these reasons, we offer first-time students additional/extended support to help them navigate the Unisa teaching and learning journey seamlessly and with little difficulty and few barriers.

Unisa's First-Year Experience (FYE) Programme has been specially designed to provide you with prompt and helpful information about the services that the institution offers.

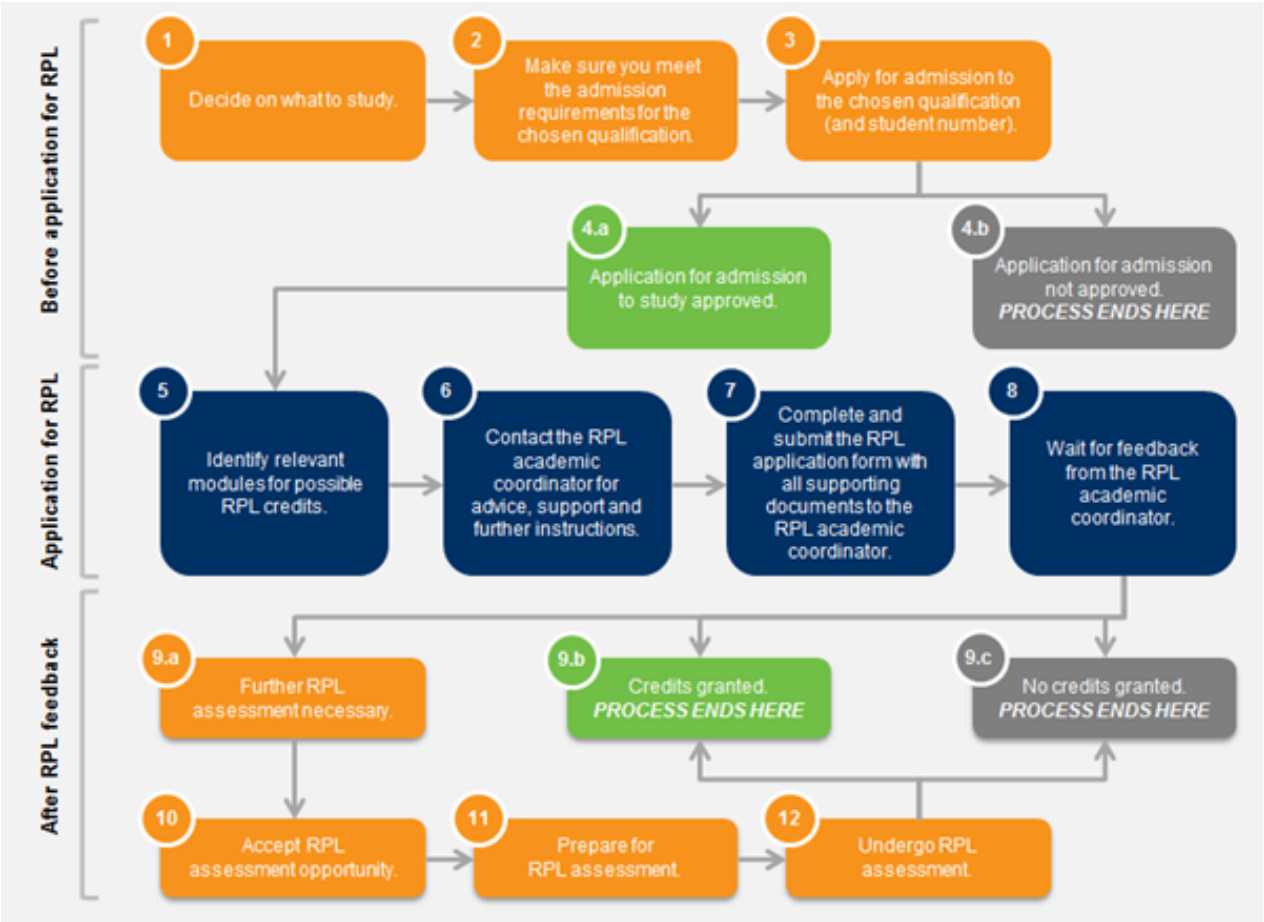
The following FYE services are currently available:



To ensure that you do not miss out on important academic and support communication from the SRU, please check your myLife inbox regularly.

6.3 Using Recognition of Prior Learning (RPL) to apply for module credit within a qualification

Now that you are a registered student, you are advised to familiarise yourself with the learning outcomes of the module or modules you have chosen. If you have been exposed to those learning outcomes for three years or more – either through work experience or other involvement – you can apply to be exempted from completing assignments and writing examinations. As part of your application for this exemption, you will be required to compile a portfolio of evidence substantiating how your experience is equivalent to the learning outcomes. The diagram below shows the steps involved in obtaining recognition of prior learning (RPL) for module credit. For more information on the process, RPL fees, and the contact details of your college RPL coordinator, visit the Unisa website: www.unisa.ac.za/rpl.



6.4 Compulsory Completion of the Academic Integrity Course

Students registered for NQF 5 – 8 programmes are required to complete the Academic Integrity Course annually.

Academic integrity represents our commitment to and demonstration of honest and ethical behaviour in academic settings. Academic integrity is a foundational principle at Unisa, underpinning the quality and credibility of our qualifications. At its core, the Academic Integrity Course is designed to advance values such as discipline, fairness, honesty and commitment, which are important to your success as a student and your future as a professional.

The course can be accessed at: https://mooc.unisa.ac.za/. Use your myLife-credentials to log in.

For new students

If you are a new student enrolling for the first time in 2026 , you must complete the full Academic Integrity Course as part of your orientation. This course is designed to instill academic values and equip you with the necessary skills required to uphold integrity throughout your academic journey.

The course comprises five comprehensive study units:

1. Unisa's values and mission on academic integrity principles.
2. Defining academic integrity in an open distance learning (ODL) environment.
3. Basic skills in academic writing.
4. Ethical usage of artificial intelligence.
5. Unisa's processes in identifying academic misconduct and detection tools.

For returning students

Those of you who completed the Academic Integrity Course in 2025 are required to complete the refresher version of the course in 2026. This course serves to reinforce the principles and practices of academic integrity. While it is a shortened version of your previously completed Academic Integrity Course, it remains compulsory and must be completed within the specified period.

The refresher course aims to:

- Reinforce understanding of academic integrity principles.
- Update you on any changes in policies or practices.
- Ensure continued commitment to ethical academic conduct.

For both new and returning students, it is important to complete the respective versions of the Academic Integrity Course within the specified periods. Unisa remains steadfast in its commitment to fostering a culture of honesty, fairness and responsibility. Non-compliance may lead to academic consequences as outlined in Unisa's Student Rules. For support, contact mymodule22@unisa.ac.za.

7 STUDY PLAN

Your study plan of the module is outlined below. Please refer to the general management and planning skills guidelines in the *Study@Unisa* brochure for further details.

The study plan below shows the content to be covered during specific periods of the year in terms of the broad concepts or topics, the study guide units and the prescribed book chapters. Your studies will be largely guided by the tutorial discussions and learning activities, and the assignments, which are all based on the same study plan. You should therefore participate as much as possible in the tutorial discussions and complete assignments and the learning or self-assessment activities linked to each topic in order to do well in the assignments, and for you to be well prepared for the final examination.

NB Note that Assignment 1 is the compulsory assignment and it might include other part of the units in the prescribed book or the study guide, please don't be surprised to see questions from different units.

Also note the following study plan is provided as a guide to help you organise your time effectively over the current academic year. While it sets out a recommended pace for covering the course content, it is important that you adapt it according to your broader academic and work schedule. For instance, if you are registered for additional modules or balancing other responsibilities, be sure to plan realistically while keeping up with the weekly expectations of this course.

Study Plan and Guidelines

MAT2612 requires consistent practice and regular problem-solving alongside a solid understanding of concepts. Set aside at least five hours per week for focused study, including reading, working through examples, and reviewing past material. Begin each topic by reading the relevant sections of the textbook and study guide, then immediately apply what you've learned by solving problems. Don't rely on passive reading — mathematical understanding comes through active problem-solving and repeated practice.

Use the self-assessment tasks and their solutions posted on myModules to test your progress and identify areas that need more attention. Try a range of problems, not just those similar to worked examples, and make a habit of drawing key diagrams such as Hasse diagrams, digraphs, and Karnaugh maps to reinforce visual understanding.

Start assignments early to give yourself time to tackle difficult questions and revisit key concepts. Once assignments are marked, carefully review the memorandums and use your mistakes as learning opportunities. Keep a notebook or digital journal of solutions, corrections, and summary notes.

March – Mid-May (Weeks 1–12)

The goal is to build a solid mathematical foundation. Read through Chapter 1 of the prescribed textbook (*Discrete Mathematical Structures* by B. Kolman, R.C. Busby & S. Ross). Focus on sets, Operations on Sets, Sequences, Properties of integers and Matrices. Then, turn to Chapters 2 and 3, which introduce logic and counting techniques.

Pay particular attention to the structure of logical statements. Understand conditionals, converses, negations and logical equivalence. Learn how to construct formal mathematical arguments using direct proof, proof by contradiction and especially Mathematical induction — a method that you will apply throughout the course, including in the examination.

Once you're comfortable with logic, move on to counting methods. Learn when to apply the sum and product rules, and how to work with permutations and combinations, including cases with and without repetition. Understand the Pigeonhole principle and how it can be used to prove the existence of certain configurations. Assignment 1 will cover the first three chapters.
Expected due date for Assignment 1 is mid-May

Mid-May – Mid-June (Weeks 13–16)

Next, read through Chapter 4 (Relations and digraphs). Explore how elements in a set relate to one another. Study key properties such as reflexivity, symmetry, antisymmetry, and transitivity. Learn to represent relations using matrices and directed graphs (digraphs).

You will also learn to apply Warshall's algorithm to compute the transitive closure of a relation and to classify equivalence relations by examining how they partition a set. Draw digraphs and build matrices by hand to reinforce your understanding of each relation.
Expected due date for Assignment 2 is mid-June

Mid-June – Mid-July (Weeks 17–20)

Now, focus on Chapter 5 (Functions), which deals with functions and permutations. Study the different types of functions, i.e., one-to-one (injective), onto (surjective), and bijective. Understand what makes a function invertible and how to find its inverse. Learn about Big-O notation, which is used to describe the growth rates of functions.

You'll then study permutations. That is, how to represent arrangements of elements using cycle notation, how to determine whether a permutation is even or odd (parity), and how to break down permutations into disjoint cycles.

Practice writing and simplifying permutations manually — this helps build both speed and understanding.
Expected due date for Assignment 3 is mid-July

Mid-July – Mid-August (Weeks 21–24)

And finally, read through Chapter 6 (Order Relations and Structures), which introduces ordered structures and logic-based algebra. Study partially ordered sets and learn how to draw Hasse diagrams. Be able to identify *greatest, least, maximal and minimal elements* within a set. Learn how to recognise and describe lattices and Boolean algebras. You will also work with Boolean functions, construct truth tables and simplify logical expressions using Karnaugh maps.
Expected due date for Assignment 4 mid-August

Mid-August – September (Weeks 25–28)

At this point, you've completed the syllabus. Use this period to review all topics by redoing assignment problems, working through past examination papers which are posted on myModules (no memorandum will be provided) and Self-assessment tasks. Pay close attention to recurring themes or concepts you found difficult.

October – November

You will write the final examination during this period. The final exam is a 2-hour Online examination paper that covers all five chapters of the prescribed textbook. The Study guide will give details on which parts of the prescribed textbook to focus on. Expect a balance of definitions, structured proofs and applications.

8 HOW TO STUDY ONLINE

What does it mean to study fully online?

Studying fully online modules differs completely from studying some of your other modules at Unisa.

- **All your study material and learning activities for online modules are designed to be delivered online on *my*Unisa.**
- **All your assignments (assessments) must be submitted online**. This means that you will do all your activities and submit all your assignments on myUnisa. In other words, you do **NOT** post your assignments to Unisa using the South African Post Office. You do **NOT** send assignments by email as such will not be considered for marking or a zero mark will be awarded.
- **All communication between you and the University happens online.** Lecturers will communicate with you via e-mail and **Chats**, **Blogs**, and use the **Announcements**, the **FAQs**, the **Discussion Forums** and the **Questions and Answers** tools. You can also use all of these platforms to ask questions and contact your lecturers.

9 ASSESSMENT

9.1 Assessment criteria

There are four written assignments and one examination for this module.

Examination admission
Please note that lecturers are not responsible for examination admission, and ALL enquiries about examination admission should be directed by e-mail to exams@unisa.ac.za.

You will be admitted to the examination if you obtain a subminimum of 40% on your year mark

Note that your marks for the assignments contribute 20% to your final mark (the remaining 80% is contributed by the final examinations).

9.2 Assessment plan

Assignments are seen as part of the learning material and assessment for this module. As you do the assignments, discuss with fellow students or tutor, you are actively engaged in learning. It is therefore important that you complete all the assignments. For each of the four assignments you should answer every question. Assignment 01 is compulsory for exam admission. All four assignments count towards your year mark for MAT2612, and the year mark is the average of all the assignments. Your year mark will contribute 20% and your exam mark 80% to the final mark. With respect to assignments, we would like to emphasize the following points:

- All assignments for MAT2612 must be **handwritten**. Do not use *Word* or LaTeX to write your assignments; and all typed assignments will receive a zero mark. You are however, allowed to submit digitally handwritten scripts.
- Please take heed of the closing dates for the assignments. In order to obtain full credit for your work, you must see to it that your assignments are submitted or before the closing dates. You will receive the solution for both assignments, only if you submit the relevant assignment. No solution will be posted to you if the relevant assignment is not received. These solutions will be available about two weeks after the closing date of the relevant assignment. If there is an extension by the University for whatever reason, the solutions will be released a week after the extended date.
- The work you submit for marking should be your best version, an assignment should not just be neat, but also logical. An example of sloppy presentation is to simply write down a series of statements without any indication of connection between them.
- The semester mark for MAT2612 in 2026 is 20% of the average mark obtained in Assignment 01 to Assignment 04. Your final mark will be the sum of your semester mark and 80% of the exam mark. Please apply the following formula to calculate your final mark, where AS1, AS2,

AS3 and AS4 are marks (in percentage) obtained in your Assignments, EX is your exam mark (in percentage), and FM your final mark (in percentage):

$$FM = \frac{AS1 + AS2 + AS3 + AS4}{5} + \frac{4 * EX}{5}.$$

Please, see also section 9.7 on how the examination system works.

These minimum requirements must not be regarded as sufficient preparation for success in the examination. It is important that you complete all the assignments.

NB: The assignments are not provided in this tutorial letter. Instead, the assignments will be uploaded on myUnisa. You will see and access them when you go online.

9.3 Assessment/assignment due dates

There are no assignment **due dates** included in this tutorial letter. Assignment due dates will be made available to you on the myUnisa landing page for this module. We envisage that the due dates will be available to you upon registration. Please start working on your assessments as soon as you register for the module. Log on to the myUnisa site for this module to obtain more information on the due dates for the submission of the assessments.

9.4 Submission of assessments/assignments

- Unisa, as a comprehensive open distance e-learning institution (CODeL), is moving towards becoming an online institution. You will therefore see that all your study material, assessments and engagements with your lecturer and fellow students will take place online. We use *my*Unisa as our virtual campus.

- The *my*Unisa virtual campus will offer students access to the **myModules site**, where learning material is available online and where assessments should be completed. Together, myUnisa and myModules form an online system that is used to administer, document and deliver educational material to you and to support engagement between you and Unisa's academics.

- The myUnisa platform can be accessed via https://my.unisa.ac.za. Click on the myModules 2026 button to access the online sites for the modules that you are registered for.

- The University undertakes to communicate clearly and as frequently as is necessary to ensure that you get the most out of using myUnisa. Please access the Announcements on your myModules site regularly, as this is where we will post important information.

- When you access your myModules site for the module/s you are registered for, you will see a welcome message posted by your lecturer. Below the welcome message you will see the assessment shells for the assessments that you need to complete. Some assessments may be multiple choice, some may be tests or written assessments/assignments, others may be forum discussions and so on. All assessments must be completed on the assessment shells available on the respective module platforms.

- This module has written assessments only. Please take note of the due dates by which the assessment must be submitted. Ensure that you follow the guidelines given by your lecturer to complete the assessment. Click on the submission button on the relevant assessment shell on myModules. You will then be able to upload your written assessment on the myModules site of the modules that you are registered for. Before you finalise the upload, double check that you have selected the correct file for upload. Remember, no marks can be allocated for incorrectly submitted assessments/assignments.

You need only submit your assignments electronically via *myUnisa*. Assignments can **not** be submitted by fax or e–mail nor by post as such will not be considered for marking.

Types of assignments and descriptions

All assignments are defined as either optional, mandatory, compulsory, or elective.

- **Elective assignments**
  - If you do not submit this assignment, you get no mark for it.
  - Only the best results of the required submissions will count towards your year mark.
- **Mandatory assignments**
  - The mark for this assignment contributes to the year mark.
  - If you do not submit a mandatory assignment, no mark is awarded and the year mark is calculated accordingly. You will therefore forfeit the marks attached to this assignment when the final mark for the module is calculated.
- **Compulsory assignments**
  - If you do not submit a compulsory assignment, the result on your academic record will be *absent*.
- **Optional assignments**
  - You are encouraged as a student to do optional assignment so that it may benefit your learning.

9.5 The assessments/assignments

As indicated in section 9.2, you need to complete four assessments/assignments for this module. Details of the tasks set will appear on the assessments/assignments themselves. **There are no assignments included in this tutorial letter**. Assignments and due dates will be made available to you on myModules for this module. We envisage that the due dates will be available to you upon registration. Solutions will be available on *myUnisa* under Additional Resources before the examination date.

9.6 Other assessment methods

There are no other assessment methods for this module.

9.7 The examination

Examination information and details on the format of the examination will be made available to you online via the myUnisa site. Look out for information that will be shared with you by your lecturer and for communication from the University (Examination department).

If you are registered for this module in 2026 then you will write the examination in **October/November** 2026. Please note:

- The exam is a two hour written examination.
- The use of a pocket calculator is not permitted during examination. You are **NOT** allowed to use a calculator during the exam.

The examination questions will be similar to the questions asked in the study guide and in the assignments.

If you are registered for this module in 2026 then you will write the supplementary examination in the **January/February 2027** period. During the course of the year, the Examination Section will provide you with information regarding the examination in general, examination websites, examination dates and examination times and that including the supplementary examination.

9.8 Invigilation/proctoring

Since 2020, Unisa has been conducting all its assessments online. Given the stringent requirements imposed by professional bodies, as well as increased solicitation of Unisa's students by third parties to unlawfully assist them with the completion of assignments and examinations, the University is obliged to assure the integrity of its assessment by using various proctoring tools: Turnitin, Moodle Proctoring, The Invigilator app and IRIS. These tools authenticate your identity and flag suspicious behaviour to assure the credibility of your responses during assessments. The description below is for your benefit as you may encounter any or all of these in your registered modules:

Turnitin is a plagiarism software that facilitates checks for originality in students' submissions against internal and external sources. Turnitin assists in identifying academic fraud and ghost writing. Students are expected to submit **typed** responses for utilisation of the Turnitin software.

The **Moodle Proctoring** tool is a facial recognition software that authenticates students' identity during their Quiz assessments. This tool requires access to a student's **mobile or laptop camera**. You must ensure that your camera is activated in your browser settings prior to starting your assessments.

The Invigilator App is a mobile application-based service that verifies the identity of an assessment participant. The Invigilator app detects student dishonesty-by-proxy and ensures that the assessment participant is the student registered for the module concerned. This invigilation tool requires students to download the app from the Google Play Store (Android devices), the Huawei AppGallery (Huawei devices) or the Apple App Store (Apple devices) on their **camera-enabled** mobile devices prior to their assessment.

IRIS Invigilation software verifies the identity of a student during assessment and provides for both manual and automated facial verification. It has the ability to record and review a student's assessment session. It flags suspicious behaviour by the students for review by an academic administrator. IRIS software requires installation on students' laptop devices that are enabled with a webcam.

Students who are identified and flagged for suspicious dishonest behaviour arising from the invigilation and proctoring reports are referred to the disciplinary office for formal proceeding.

Please note:
Refer to your module assessment information on the myModules sites to determine which proctoring or invigilation tool will be used for your formative and summative assessments.

10 ACADEMIC DISHONESTY

10.1 Plagiarism

Plagiarism is the act of taking the words, ideas and thoughts of others and presenting them as your own. It is a form of theft. Plagiarism includes the following forms of academic dishonesty:

- Copying and pasting from any source without acknowledging the source.
- Not including references or deliberately inserting incorrect bibliographic information.
- Paraphrasing without acknowledging the original source of the information.

10.2 Cheating

Cheating includes, but is not limited to, the following:

- Completing assessments on behalf of another student, copying the work of another student during an assessment, or allowing another student to copy your work.
- Using social media (e.g. WhatsApp, Telegram) or other platforms to disseminate assessment information.
- Submitting corrupt or irrelevant files, this forms part of examination guidelines.
- Buying completed answers from so-called ”tutors” or internet sites (contract cheating).

For more information about plagiarism, follow the link below:

https://www.unisa.ac.za/sites/myunisa/default/Study-@-Unisa/Student-values-and-rules

11 STUDENTS LIVING WITH DISABILITIES

The Advocacy and Resource Centre for Students with Disabilities (ARCSWiD) provides an opportunity for staff to interact with first-time and returning students with disabilities.

If you are a student with a disability and would like additional support or need additional time for assessments, you are invited to contact (Mr M Kgarose @ kgarom@unisa.ac.za) to discuss the assistance that you need.

12 FREQUENTLY ASKED QUESTIONS

The *Study@Unisa* brochure contains an A-Z guide of the most relevant study information.

13 SOURCES CONSULTED

The Study Guide and the prescribed textbook were consulted in preparing this tutorial letter.

14 IN CLOSING

Do not hesitate to contact us by e-mail if you are experiencing problems with the content of this tutorial letter or with any academic aspect of the module.

We wish you a fascinating and satisfying journey through the learning material, and trust that you will complete the module successfully.

Enjoy the journey!

MAT2612 lecturers
Department of Mathematical Sciences

ADDENDUM

Note that it is crucial to understand the content of this module in order to be able to properly do your assignments on your own and solve related problems. For this purpose, video notes will be uploaded under Additional Resources of the main module. In order to fully understand the concepts of this module and benefit from this course, you can request a MS Teams meeting for live discussions with the lecturer where recordings and video notes can be made available to your fellow students.

Note that the lecturers for this module will provide live discussions (students engagement and interactions) on Teams about the concepts in order to facilitate the understanding for this module.

Note that Feedback on Assignments may be discussed live on MS Teams with the lecturers for this module. The live discussion will be interactive between the lecturers and the students.

ANNEXURE: GLOSSARY OF TERMS

The Glossary of terms will be uploaded online only.

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